

INDUSTRIAL TRAINING REPORT

INFORMATION TECHNOLOGY

Submitted in partial fulfillment of the Requirements for the award of
Degree of Bachelor of Engineering in Information Technology



SUBMITTED BY :

NAME : Solanki Savan Govindbhai

ENROLLMENT NO. : 20BECE54059

SEMESTER : 8TH / BRANCH : CE.

SUBMITTED TO:

Department of Information Technology

Vidush Somany Institute of Technology & Research, Kadi

Kadi Sarva Vishwavidyalaya University, Gandhinagar

A)INTERNSHIP CERTIFICATE



F6, Dhanlaxmi Chambers, Near Sarvodaya Co op bank,
Ashram road Ahmedabad-380014

Completion Certificate

This is to certify that, **Mr. Solanki Savan Govindbhai (20BECE54059)** Student of **VSITR – Gandhinagar**, has successfully completed a 15 Days Internship in the field of **Python** during the period of **1st January 2024 to 1st February 2024**.

During the period of this internship program with us, student of your College/University exposed to different processes and found sincere and hardworking.

BRAINYBEAM INFO-TECH PVT. LTD.
AHMEDABAD

With Best Regards,
Mr. Sagar Jasani
CEO.
BrainyBeam Info-Tech Pvt. Ltd
Ahmedabad (Gujarat-INDIA)



+91 90332 37336



sagar@brainybeaminfotech.com

B) DECLARATION

I hereby declare that the Industrial Training Report on Name of Department Section, Industry is an authentic record of my own work on requirement of Minor/Major Industrial Training during the period from 1th Jan 2024 to 1th Feb 2024 for the award of degree of B.E(IT)/Vidush Sonam Institute of Technology & Research, Kadi, Kadi Under the guidance of **Prof. Anand Sutariya.**

Date : _____

.....
Solanki Savan

20BECE54059

Certified that the above statement made by the student is correct to the best of our knowledge and belief.

Examined By:

Head Of Department:

Prof. Anand Sutariya

Prof. Nehal Shah

C) ACKNOWLEDGEMENT

First and foremost .I wish to express my sincere thanks and gratitude to my esteemed Mentor “**Vrunda Hingrajia**” Who has contributed to much for successful completion of my Industrial Training by his thoughtful reviews and valuable guidance

Next I would like to tender my sincere thanks to “**Prof. Nehal Shah**” for the co-operation and encouragement

.....
Solanki Savan

20BECE54059

3. ABOUT COMPANY & INDUSTRY



BrainyBeam Technologies Pvt. Ltd. in Ahmedabad & United is your top choice for professional WordPress development, website design, SEO, and digital marketing services.

At Ahmedabad, the company is located at Income tax cross Rd.

BrainyBeam, we see Innovation as a clear differentiator. Innovation, along with focus on deep, long-lasting client relationships and strong domain expertise, drives every facet of our day-to-day operations.

BrainyBeam Technologies was founded with a vision to address growing businesses' needs of reducing the time to market and cost effectiveness required to develop and maintain unique and customized web and mobile solutions. We are uniquely and strategically positioned to partner with startups and leading brands to help them expand their business and offer the most effective and cost efficient solutions that provide revenues and value to their business needs.

All processes are well documented, that enable us to deliver more effective and innovative solutions with each project we work on. We provide services that conform to customers' requirements, at a satisfactory overall cost, having regard to functionality, reliability, durability, usability, appearance and safety.

.

Overall, BrainyBeam Technologies Pvt. Ltd. is a leading service-based company that provides businesses with a comprehensive range of services to help them achieve their goals.

4. TABLE OF CONTENT

Sr No.	Table of Content	Page No.
1.	Cover Page	1
2.	Inner Page	2
	A) Certificate of Company	2
	B) Declaration by Student	3
	C) Acknowledgement	4
3.	About Company/Industry	5
4.	Table of Contents	6
5	List of Tables	7
6	List of Screenshots	8
7	Industrial report chapters	9
	A) Introduction	9
	B) Formal training provided	10
	C) Industrial Training	11
	a) Objective of training	11
	b) Tools & Technologies used	12
	D) Weekly Overview of Activities	13
8	Project Experience / Coding	16
9	Conclusion	22
10	References	23

5. List Of Tables

Sr. no.	Table	Page No.
1	Week 1	13
2	Week 2	13
3	Week 3	14
4	Week 4	14
5	Week 5	15

6. List of Screenshots

Sr. no.	Table	Page No.
1	Fig-1	18
2	Fig-2	18
3	Fig-3	19
4	Fig-4	19
5	Fig-5	20
6	Fig-6	20
7	Fig-7	21
8	Fig-8	21
9	Fig-9	21

7. Industrial Report Chapters

A) Introduction

This report documents my internship experience at **BrainyBeamTechnologies Pvt. Ltd.** As a **Python with Django Intern.**

During the period from Jan 1th to Feb 1th, 2024, I had the privilege of undertaking an enriching internship experience at BrainyBeamTechnologies Pvt Ltd as a Python with django Intern. This opportunity provided me with invaluable insights into the practical applications of python django techniques within a dynamic and innovative technological environment. Throughout the internship, I had the chance to collaborate with talented professionals, tackle real-world challenges, and contribute to impactful projects that exemplified Maxgen Technologies' commitment to excellence and innovation in the field of artificial intelligence. This report aims to reflect on my experiences, accomplishments, and learnings during my tenure at BrainyBeamTechnologies, highlighting the valuable skills and knowledge gained through hands-on engagement in python with django projects.

Duties and Responsibilities:

As a python with django Intern at BrainyBeamTechnologies, my duties included data preprocessing, model development, and performance evaluation using Python and django libraries.

Conducting experiments, analyzing results, and fine-tuning models to improve performance and accuracy.

Providing support in troubleshooting and debugging machine learning models and algorithms to resolve issues effectively.

These responsibilities collectively contributed to my growth as a python with django practitioner and enriched my understanding of practical applications in a professional setting at BrainyBeam Technologies.

B) Formal Training provided

This section provides an overview of the training provided by the college for the internship program. The Internship program is an integral part of the curriculum, which provided students with an opportunity to gain practical experience in their respective fields. The college provides training to students to ensure that they are prepared to make the most of the internship program.

The formal training provided by my college during the sixth semester complemented my internship experience in python with django at BrainyBeam Technologies. The college curriculum included structured classroom sessions, workshops, and practical exercises tailored to enhance understanding and proficiency in machine learning concepts. This also includes :

Communication skills

Professional skills

Technical skills

Industry insights

Continuous assessment

C) Industrial Training

a) Objective of Training :

The objective of the industrial training component of my internship at BrainyBeamTechnologies was to bridge the gap between theoretical knowledge gained in academic coursework and practical skills required in a professional environment. Through hands-on experience and exposure to real-world projects, the training aimed to:

1. Enhance proficiency in applying django concepts, algorithms, and methodologies to solve complex problems encountered in industry settings.
2. Acquire practical skills in data preprocessing, feature engineering, model development, and evaluation techniques essential for building robust and effective django models.
3. Gain insight into industry practices, standards, and workflows pertaining to django projects, including version control, documentation, and collaboration within a team environment.
4. Develop professional competencies such as critical thinking, problem-solving, communication, and project management skills necessary for success in the field of python programing.

By aligning the objectives of the industrial training with the practical requirements of the industry, the internship aimed to equip me with the knowledge, skills, and experience necessary to transition seamlessly from academia to the professional world of python programming.

b) Tools & Technology Used

During my internship at BrainyBeamTechnologies, I utilized a range of tools and technologies to conduct python with django experiments, analyze data, and develop models. The primary tools and technologies employed include:

1. **Python:** Python served as the primary programming language for implementing machine learning algorithms and data analysis tasks. Its versatility, rich ecosystem of libraries, and ease of use made it an ideal choice for developing machine learning solutions.
2. **Jupyter Notebook:** Jupyter Notebook was the preferred code editor for writing and executing Python code. Its interactive environment facilitated exploratory data analysis, iterative development, and collaborative work on machine learning projects.
3. **Django:** Explain the role of Django as a high-level Python web framework for rapid development of secure and maintainable websites and web applications.
4. **Integration of Python and Django:** Highlight how Python and Django complement each other in web development projects.

By leveraging these tools and technologies, I was able to effectively implement django learning algorithms, analyze datasets, visualize results, and derive insights to address the objectives of various projects undertaken during my internship at BrainyBeam Technologies.

D) Weekly Overview of Activities

1 st WEEK		
DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
01/01/24	Monday	Introduction to company
02/01/24	Tuesday	Python Tutorial
03/01/24	Wednesday	Django Basics: Setting up a Django Project
04/01/24	Thursday	Django Models and ORM (Object-Relational Mapping)
05/01/24	Friday	Assessment: Implementing Django Models and Data Management

2 nd WEEK		
DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
08/01/24	Monday	Django Views and URL Routing
09/01/24	Tuesday	Django Templates and Frontend Development
10/01/24	Wednesday	Django Forms and Form Handling
11/01/24	Thursday	Django Authentication and User Management
12/01/24	Friday	Assessment: Implementing Django Views, Templates, and Forms

3 rd WEEK		
DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
15/01/24	Monday	Django Admin Interface and Customization
16/01/24	Tuesday	Django Middleware and Request-Response Cycle
17/01/24	Wednesday	Django REST Framework and API Development
18/01/24	Thursday	Deployment of Django Applications
19/01/24	Friday	Assessment: Implementing Django REST APIs and Deployment

4 th WEEK		
DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
22/01/24	Monday	Testing in Django: Unit Testing and Integration Testing
23/01/24	Tuesday	Django Security Best Practices
24/01/24	Wednesday	Working with Django Packages and Libraries
25/01/24	Thursday	Django Performance Optimization Techniques
26/01/24	Friday	Advanced Django Concepts: Class-Based Views

5th WEEK		
DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
29/01/24	Monday	Project – Online Bookstore
30/01/24	Wednesday	Project Presentation
31/01/24	Thursday	Reflection and feedback session / Wrap-up and conclusion

8. Project Experience / Coding

Online Bookstore

The Online Bookstore project undertaken during my internship at BrainyBeam Technologies involved developing a python django to show books based on various factors. The objective of the project was to build an accurate and reliable predictive model that could assist readers, Project Experience / Coding

Project Objectives:

1. **Data Collection and Preprocessing:** Gather book data from various sources including publishers, APIs, and databases.Cleanse and preprocess the data to handle inconsistencies, ensuring accurate and reliable information for users.
2. **Feature Engineering:**Identify key features influencing book sales and user preferences such as genre, author, publication date, ratings, and reviews.Engineer features to enhance user experience, such as personalized recommendations based on browsing history and purchase patterns.
3. **Model Development:** Explore Django's built-in features and extensions to develop a robust backend for the online bookstore.Implement core functionalities including user authentication, book catalog management, shopping cart, and checkout process.
4. **Model Evaluation:** Evaluate system performance and usability through user testing and feedback gathering.Validate the reliability and efficiency of the bookstore's functionalities, ensuring seamless user interactions and transaction processing.
5. **Deployment and Integration:** Deploy the online bookstore application on a production server, ensuring scalability and reliability.Integrate payment gateways and shipping services to enable secure transactions and timely delivery of purchased books.

Project Deliverables:

1. **Functional Online Bookstore Application:** A fully developed and functional online bookstore application built using Python with Django framework. The application will include features such as user authentication, book catalog management, search and filtering options, shopping cart functionality, secure checkout process, and integration with payment gateways.
- 2 **Documentation Package:** A comprehensive documentation package providing insights into the project objectives, methodology, data sources, feature selection criteria, Django application structure, and deployment instructions. The documentation will also include detailed explanations of each feature implemented in the application, along with screenshots and code snippets for reference.
3. **User Manual:** A user manual outlining the usage instructions for the online bookstore application. The manual will provide step-by-step guidance on how to navigate the application, search for books, add them to the cart, proceed to checkout, and make purchases securely. It will also include troubleshooting tips and FAQs for common user queries.

4. **Presentation Slide Deck:** A visually engaging presentation slide deck summarizing key aspects of the project, including the project workflow, challenges encountered, lessons learned, and future enhancements. The presentation will highlight the features of the online bookstore application, user interface design choices, and integration with external services for payment processing and order fulfillment.

Project Outcomes:

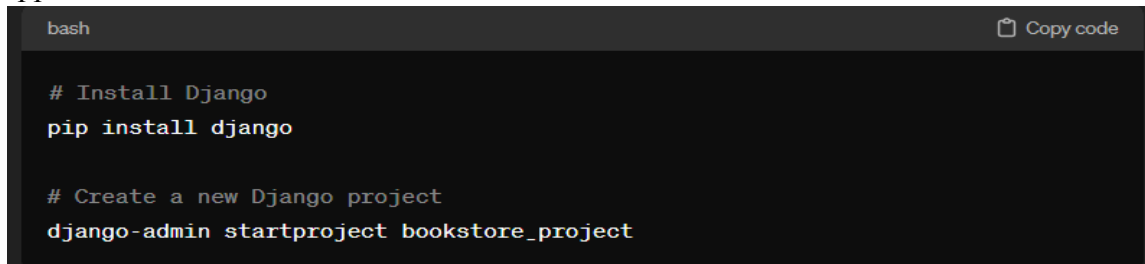
The project outcomes for the online bookstore using Python with Django encompass successful deployment of a seamless shopping experience, enhanced user engagement, secure transactions, improved sales and revenue generation, scalability, positive user feedback, valuable insights, and competitive advantage. The application is effectively deployed, providing users with an intuitive interface for browsing, searching, and purchasing books securely. Personalized recommendations and user-friendly features contribute to increased user engagement and conversion rates, leading to improved sales and revenue. The scalable architecture allows for future expansion and customization to meet evolving business needs. Positive user feedback and valuable insights gleaned from data analytics inform strategic decision-making, while the successful implementation establishes the business as a competitive player in the e-commerce market.

Tools and Technologies used for Project

Python
Django Framework
HTML
CSS
JavaScript
Bootstrap Framework
SQLite
PostgreSQL
Git
Payment Gateway Integration
Deployment Platforms (e.g., Heroku, AWS, DigitalOcean)
Django Packages and Libraries

Coding

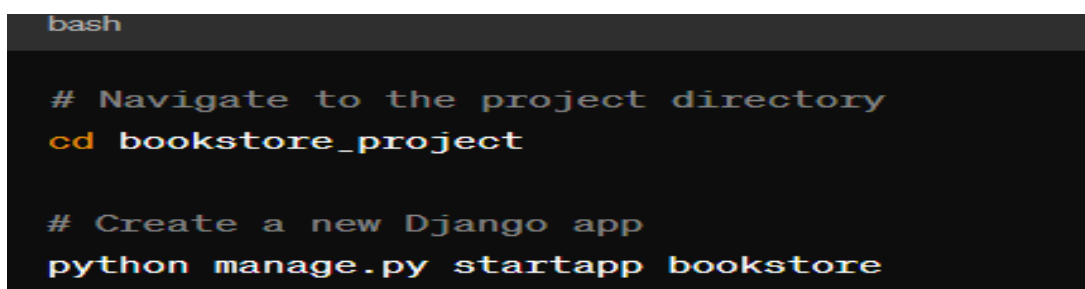
Setting Up Django Environment: The initial step involves setting up a Django environment by installing Django and creating a new Django project using the command line interface. This establishes the foundation for developing the online bookstore application..

A terminal window with a dark background. The title bar says 'bash' and there is a 'Copy code' button in the top right corner. The terminal contains two lines of code: a comment '# Install Django' followed by the command 'pip install django', and another comment '# Create a new Django project' followed by the command 'django-admin startproject bookstore_project'.

```
bash  
  
# Install Django  
pip install django  
  
# Create a new Django project  
django-admin startproject bookstore_project
```

Fig - 1

Creating Django App: Next, a Django app specifically for the online bookstore is created within the Django project. This app will contain the necessary models, views, templates, and other components required for building the bookstore functionality.

A terminal window with a dark background. The title bar says 'bash'. The terminal contains two lines of code: a comment '# Navigate to the project directory' followed by the command 'cd bookstore_project', and another comment '# Create a new Django app' followed by the command 'python manage.py startapp bookstore'.

```
bash  
  
# Navigate to the project directory  
cd bookstore_project  
  
# Create a new Django app  
python manage.py startapp bookstore
```

Fig-2

Defining Models: Django models are defined to represent the core entities of the online bookstore, such as books, authors, genres, users, and orders. Each model class corresponds to a database table, defining the fields and relationships **between entities**.

```
# In models.py of the bookstore app

from django.db import models

class Author(models.Model):
    name = models.CharField(max_length=100)

class Genre(models.Model):
    name = models.CharField(max_length=100)

class Book(models.Model):
    title = models.CharField(max_length=200)
    author = models.ForeignKey(Author, on_delete=models.CASCADE)
    genre = models.ForeignKey(Genre, on_delete=models.CASCADE)
    publication_date = models.DateField()
    price = models.DecimalField(max_digits=10, decimal_places=2)

class User(models.Model):
    username = models.CharField(max_length=100)
    email = models.EmailField()
    # Add other user-related fields as needed

class Order(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    book = models.ForeignKey(Book, on_delete=models.CASCADE)
    quantity = models.IntegerField()
    total_price = models.DecimalField(max_digits=10, decimal_places=2)
```

Fig -3

Implementing Views: Views are implemented to handle user requests and generate appropriate responses. Django views utilize functions or classes to process incoming HTTP requests, retrieve data from the database, and render HTML templates to display dynamic content.

```
# In views.py of the bookstore app

from django.shortcuts import render
from .models import Book

def book_list(request):
    books = Book.objects.all()
    return render(request, 'bookstore/book_list.html', {'books': books})
```

Fig - 4

Creating URLs: URL patterns are defined to map specific URLs to corresponding views within the Django app. This enables navigation within the online bookstore application and determines how URLs are processed and routed.

```
# In urls.py of the bookstore app

from django.urls import path
from . import views

urlpatterns = [
    path('books/', views.book_list, name='book_list'),
    # Add other URL patterns as needed
]
```

Fig - 5

Designing Templates: HTML templates are designed to define the structure and layout of web pages in the online bookstore application. Django's templating engine allows for dynamic content rendering, including data retrieved from the database and user interactions.

```
<!-- In book_list.html template -->

<!DOCTYPE html>
<html>
<head>
    <title>Bookstore</title>
</head>
<body>
    <h1>Book List</h1>
    <ul>
        {% for book in books %}
        <li>{{ book.title }} - {{ book.author }}</li>
        {% endfor %}
    </ul>
</body>
</html>
```

Fig - 6

Handling Forms: Django forms are utilized to handle user input, such as search queries, user registration, login, and checkout. Forms ensure data validation, security, and seamless interaction between users and the online bookstore application.

```
# In forms.py of the bookstore app

from django import forms
from .models import Order

class OrderForm(forms.ModelForm):
    class Meta:
        model = Order
        fields = ['user', 'book', 'quantity', 'total_price']
```

Fig- 7

Implementing Business Logic: Custom Python functions and methods are implemented to encapsulate the business logic of the online bookstore application. This includes functionalities such as book recommendation algorithms, order processing, and user authentication.

Testing and Debugging: The online bookstore application is thoroughly tested to ensure its functionality, reliability, and security. Unit tests, integration tests, and user acceptance tests are conducted to identify and fix any bugs or issues.

```
# Run tests
python manage.py test
```

Fig - 8

Deployment: Finally, the completed online bookstore application is deployed to a production server, making it accessible to users. Deployment involves configuring server settings, optimizing performance, and ensuring scalability and reliability of the application.

```
# Collect static files
python manage.py collectstatic

# Run server
python manage.py runserver
```

Fig - 9

9. Conclusion

My internship experience at BrainyBeam Technologies has been immensely rewarding, providing me with valuable insights, practical skills, and professional growth in the field of python using the django framework. Over the course of four weeks, I had the opportunity to immerse myself in a dynamic and innovative environment, working on diverse projects and collaborating with talented professionals.

Throughout the internship, I gained hands-on experience in data preprocessing, feature engineering, model development, and evaluation, honing my technical proficiency and problem-solving abilities. The exposure to real-world challenges and industry-standard tools such as Python, django, and Jupyter Notebook has equipped me with the practical skills necessary to tackle complex machine learning projects with confidence.

Moreover, the internship provided me with valuable mentorship, guidance, and networking opportunities, enabling me to learn from experienced professionals and expand my professional network. I am grateful for the support and encouragement received from my mentors and colleagues, who have played a pivotal role in shaping my learning journey and fostering a conducive learning environment.

In conclusion, my internship at BrainyBeam Technologies has been a transformative experience, allowing me to apply theoretical knowledge gained in academic coursework to real-world scenarios and emerging technologies. I am confident that the skills and insights gained during this internship will serve as a solid foundation for my future career endeavors in the field of data science and artificial intelligence. I am thankful for the opportunity to contribute to impactful projects, and I look forward to leveraging my learnings to make meaningful contributions to the field.

10) References

For company reference you can contact to,

[2nd, Floor, Dhanalakshmi Complex, Income Tax - Cross Road, Ashram Rd,](#)

[opp. Jet Airways Office, Sattar Taluka Society, Usmanpura, Ahmedabad,](#)

[Gujarat 380014](#) Email: contact@brainybeam.com

General inquiry :contact@brainybeam.com

Contact number : +91 9033 237 336

Technical references

Django Documentation: Official resource providing guides and references for Django web development.

Python Documentation: Comprehensive guide covering Python programming language features and libraries.

Pandas Documentation: Resourceful guide for data manipulation and analysis using pandas library.

Matplotlib Documentation: Comprehensive reference for creating visualizations with Matplotlib library.

Bootstrap Documentation: Official guide for building responsive web designs with Bootstrap CSS framework.

JavaScript MDN Web Docs: Comprehensive resource for learning JavaScript programming.

SQLAlchemy Documentation: Detailed documentation for SQL toolkit and ORM library in Python.

W3Schools: Interactive tutorials and references for web development technologies.

